Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 101606

Subject: Geotechnical Engineering – II

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) In Terzaghi's one-dimensional consolidation theory, drainage occurs
(i) Vertically only
(ii) Both directions
(iii) Horizontally only
(iv) Radially

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(b) In a triaxial test, the difference between major and minor principal stresses is known as
(i) Deviator stress
(ii) Shear stress
(iii) Normal stress
(iv) Effective stress

(c) Infinite slopes generally fail due to
(i) Toe failure
(ii) Shallow slip
(iii) Face failure
(iv) Base failure

(d) The friction circle method is used for
(i) Determining active pressure
(ii) Pile analysis
(iii) Determining critical circle
(iv) Bearing capacity

(e) In Rankine's theory, the failure surface is assumed to be
(i) Plane
(ii) Log-spiral
(iii) Circular
(iv) Parabolic

(f) Earth pressure at rest condition corresponds to
(i) No lateral strain
(ii) Passive state
(iii) Active state
(iv) Ultimate state

(g) The secondary consolidation is mainly due to
(i) Expulsion of water
(ii) Elastic rebound
(iii) Rearrangement of soil particles
(iv) None of these

(h) The failure envelope for a purely cohesive soil makes an angle of
(i) 0° with the horizontal
(ii) 90°
(iii) 45°
(iv) Equal to internal friction angle

(i) The Swedish circle method is applicable for
(i) Infinite slopes
(ii) Retaining walls
(iii) Finite slopes
(iv) Sheet piles

(j) Cone Penetration Test (CPT) measures
(i) Shear strength
(ii) Density
(iii) Resistance to penetration
(iv) Cohesion only

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Q.2	(a)	Discuss the mechanism of primary consolidation in cohesive soil with reference to Terzaghi's assumptions.	[7]
	(b)	Explain the difference between compaction and consolidation based on soil structure, drainage conditions, and time dependency.	[7]
Q.3	(a)	Describe the step-by-step procedure for conducting a laboratory consolidation test as per IS: 2720 (Part XV).	[7]
	(b)	Interpret the e–log p curve and explain the significance of compression index and recompression index.	[7]
Q.4	(a)	Explain the concept of degree of consolidation and its significance in settlement analysis.	[7]
	(b)	A clay layer of 6 m thickness is drained on both sides. Coefficient of consolidation, $Cv = 2.5 \times 10^{-3}$ cm²/sec. Determine the time required for 50% consolidation.	[7]
Q.5	(a)	Present Mohr–Coulomb failure theory and its application in determining shear strength parameters.	[7]
	(b)	Compare UU, CU, and CD triaxial tests with respect to drainage, pore-water pressure, and field conditions modeled.	[7]
Q.6	(a)	Illustrate, with neat diagrams, the stress conditions on principal planes in a soil element using Mohr's Circle.	[7]
	(b)	In a triaxial test, the cell pressure = 150 kPa and deviator stress at failure = 120 kPa. Compute the major and minor principal stresses.	[7]
Q.7	(a)	Discuss the advantages, limitations, and typical applications of the direct shear test in geotechnical engineering practice.	[7]
	(b)	Describe the vane shear test and its relevance for soft saturated clays.	[7]
Q.8	(a)	Differentiate between finite and infinite slopes with respect to failure mechanisms.	[7]
	(b)	An infinite slope in dry sand has $\varphi = 32°$ and slope angle $\beta = 24°$. Calculate the factor of safety using friction-only analysis.	[7]
Q.9	(a)	Explain the Swedish Circle (Fellenius) method used for the stability analysis of slopes.	[7]
	(b)	Describe the concept of stability numbers and the use of Taylor's charts for cohesive soil slopes.	[7]

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